PIMS PRO

Pipeline Integrity Management System

Pipeline Corrosion Assessment & Remaining Life Calculator

Version	1.0
Developer	M. Fahad Zafar
Report File	PL-002_Inspection_Report_2026-07-16.pdf
Generation Date	16 July 2026

Inspection Information

Pipe ID	PL-002
Pipeline Name	PL-002
Material	Alloy Steel
Fluid	Diesel
Inspector	Sadia Khan
Inspection Date	2026-07-16
Operating Pressure	7.0 MPa

Engineering Calculations

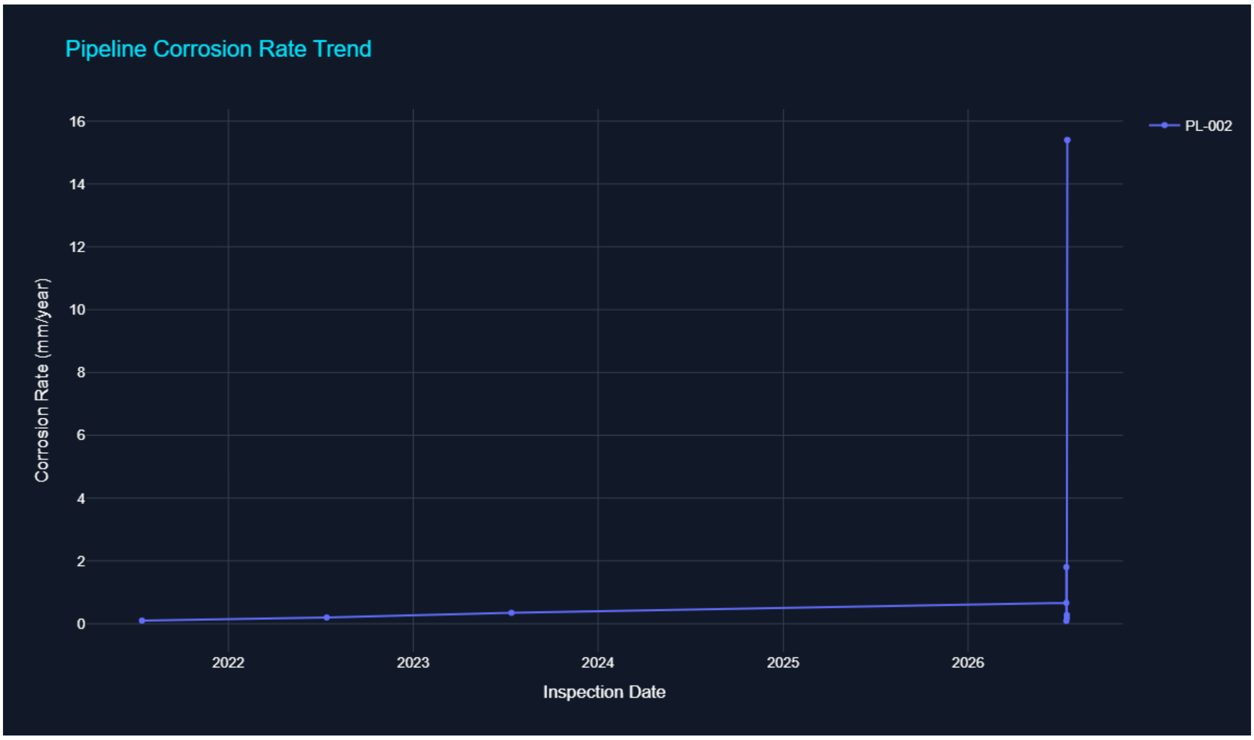
Parameter	Value

Corrosion Rate	15.40 mm/year
Minimum Thickness	19.17 mm
Remaining Life	6.74 years
Safety Factor	6.42
Corrosion Loss	38.50 %
Health Score	49.00 %
Failure Year	2033

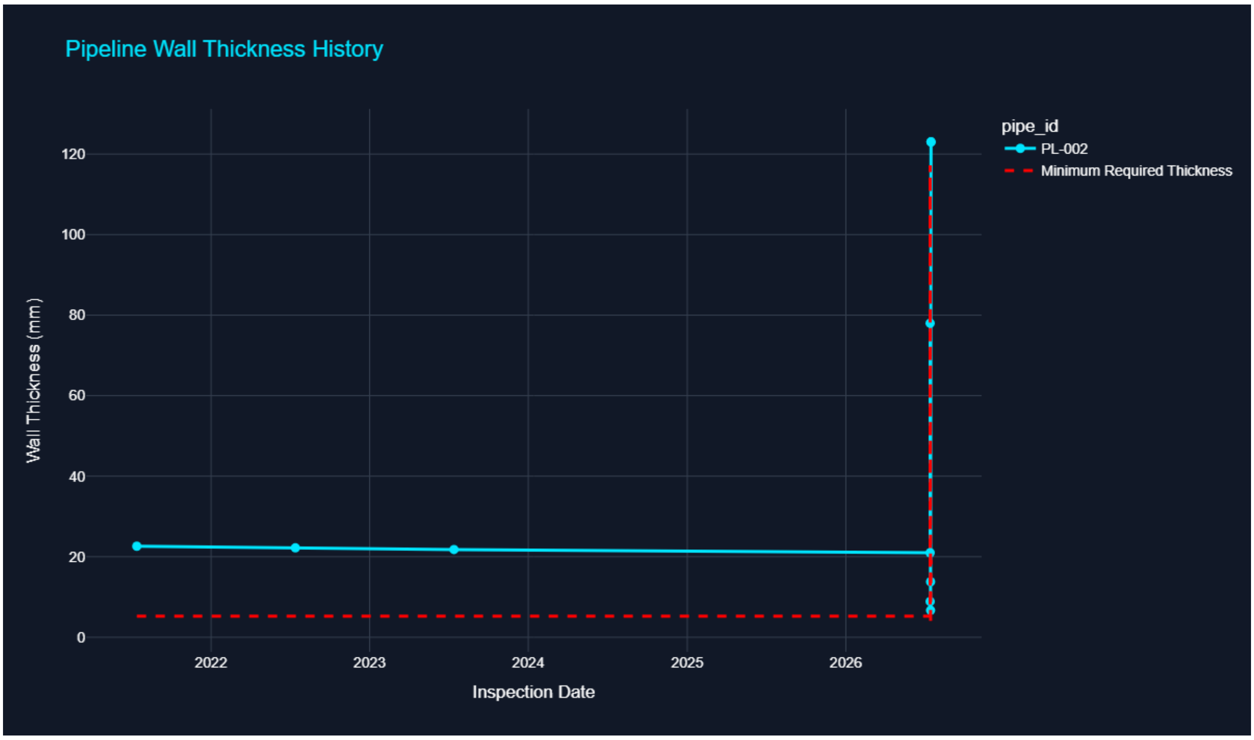
KPI Summary

Health Score	**49.00%**
Risk Level	**Critical**
Remaining Life	**6.74 Years**
Corrosion Rate	**15.40 mm/year**

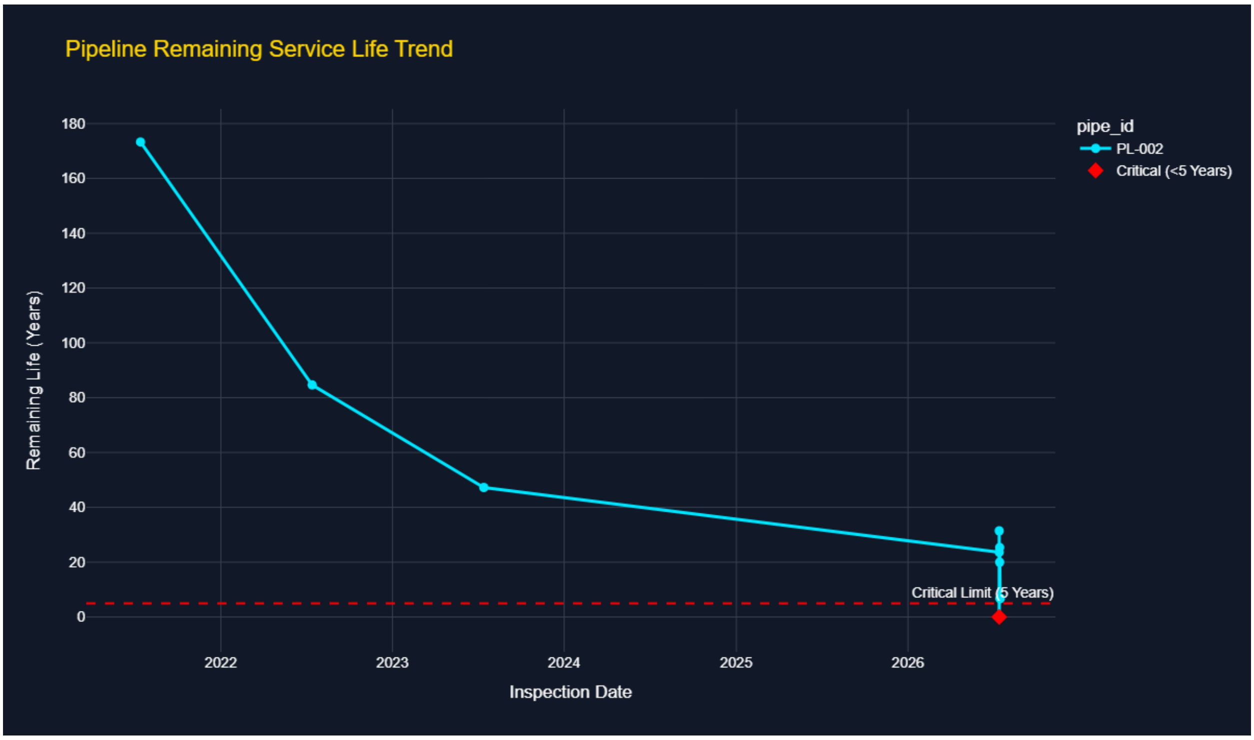
Corrosion Trend



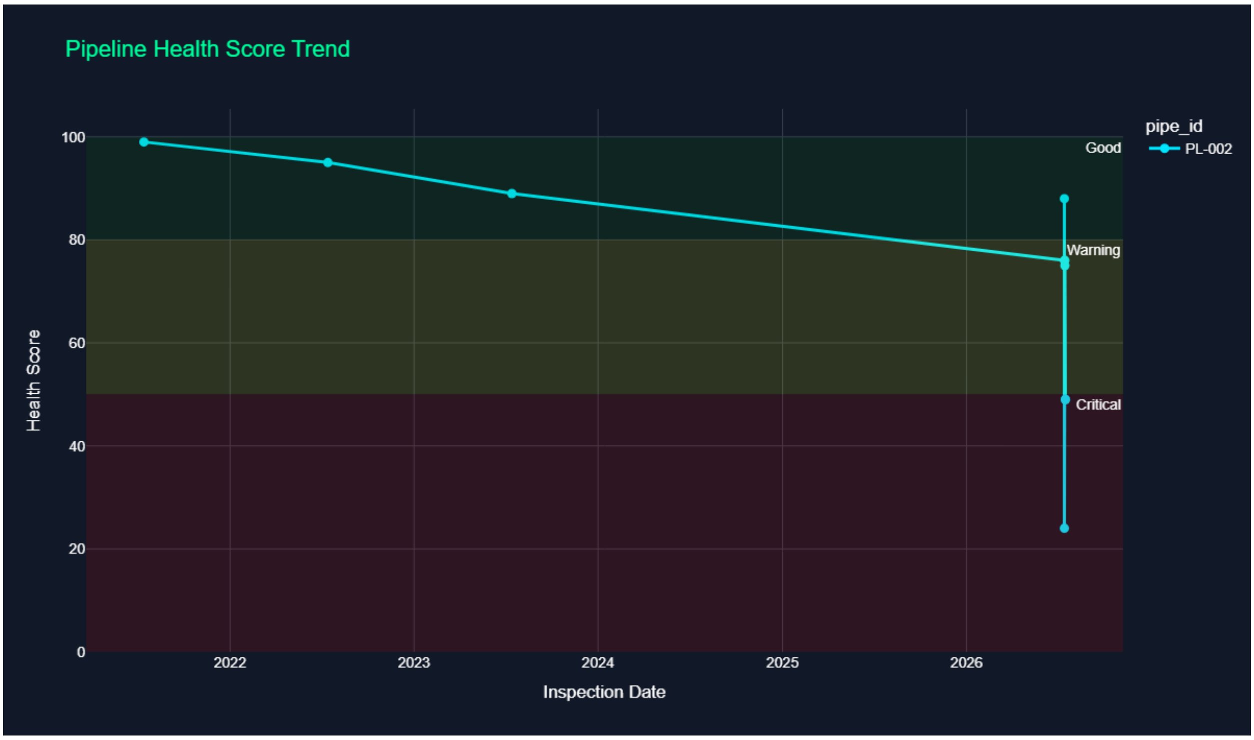
Thickness History



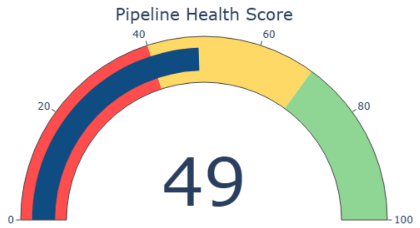
Remaining Life Trend

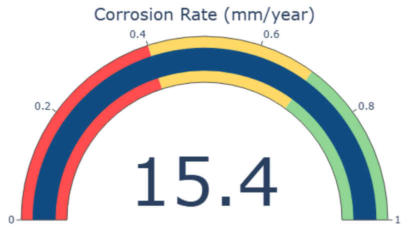


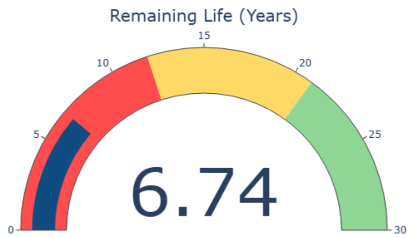
Health Trend



Engineering KPI Gauges



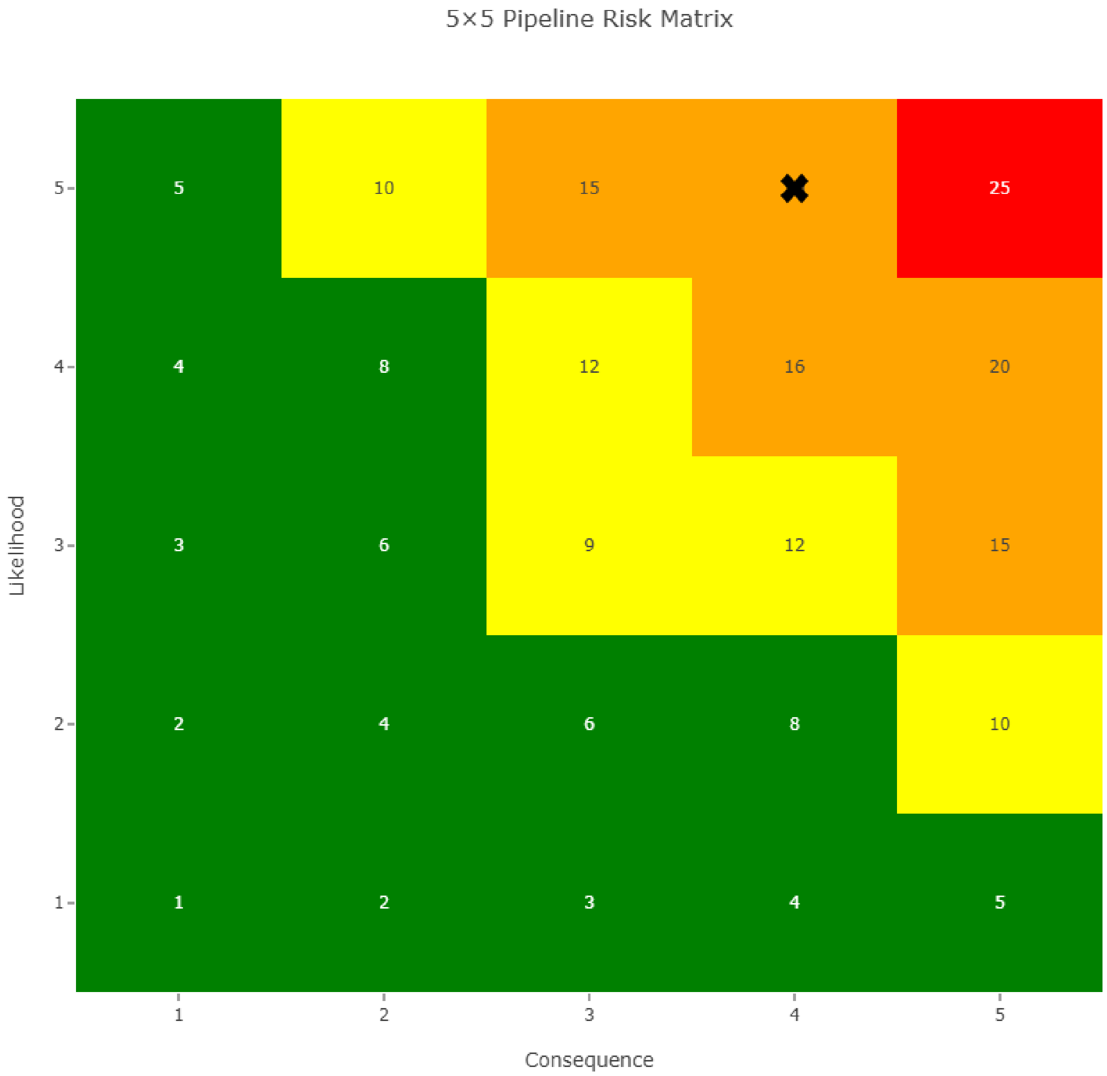




Risk Assessment

Parameter	Value
Likelihood	5
Consequence	4
Risk Score	20
Risk Category	Critical

5 × 5 Risk Matrix



Engineering Recommendation

Overall Decision	**Pipeline Requires Maintenance**
Maintenance Suggestion	**Repair or replace damaged section.**
Inspection Interval	**Inspect within 6 months.**
Priority	**HIGH**
Warning Message	**Pipeline approaching minimum thickness.**

Inspection Summary

The pipeline (PL-002) remains suitable for continued operation. The measured corrosion rate is 15.40 mm/year with an estimated remaining service life of 6.74 years. The calculated health score is 49.0% and the overall risk category is Critical. Repair or replace damaged section.

Conclusion:

Based on the inspection data and engineering calculations performed in accordance with ASME B31.3 and API 570 principles, the pipeline integrity has been evaluated and appropriate maintenance recommendations have been provided to support safe and reliable operation.

Engineering Notes

Engineer observations, maintenance remarks, or site comments may be recorded below.

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Engineering Standards

• ASME B31.3 – Process Piping

• API 570 – Piping Inspection Code

The engineering calculations, integrity assessment, corrosion evaluation, remaining life estimation, and inspection recommendations presented in this report are based on the principles and methodologies defined in ASME B31.3 and API 570. The report is intended as an engineering decision-support document and should be reviewed by qualified personnel before maintenance or operational decisions are made.